Screening Test 20/3/2021

SUBJUNIOR

1. Sadhana's date of birth is d: m: 2008 and Bhavna's date of birth is D: 05: 2008, their date of births given in the (date: month: year) format. D, d, m are all composite numbers and pairwise co-primes. Also, $D + d + m = 48$. Then, the birth month of Sadhana is

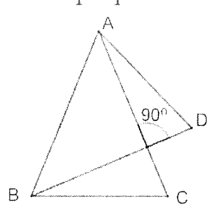
   (1) June
   (2) October
   (3) September
   (4) April

   (Question 1)

2. a, b are positive integers such that $a^2 - ab - 20b^2 - 2020 = 0$. Then, the *smallest* possible value of $a - 7b$ is

   (1) 2
   (2) −8
   (3) −222
   (4) −101
   (Question 3)

3. ABC and ABD are isosceles triangles with $AB = AC = BD$. Also, BD meets AC at
P
and is perpendicular to AC. Then $\angle C + \angle D$ is



   (1) 135°
   (2) 110°
   (3) 120°
   (4) 130°
   (Question 9)

4. The positive integer '*m*' has exactly three positive divisors and the positive integer '*n*' has exactly four positive divisors. The least possible number of positive divisors of is

   (1) 5
   (2) 6
   (3) 7
   (4) 12
   (Question 40)

5. If $x = 3$, $y = 2.3$ and $z = 3.7$ then the value of   is

(1) 1
(2)
(3)
(4) 0

(Question 34)

6. Let $a < b < c < d$ be positive integers each being less than 10. How many possible combinations of a, b, c, d are there such that LCM(a, d) = LCM(b, c)?

(1) 1
(2) 2
(3) 3
(4) 4

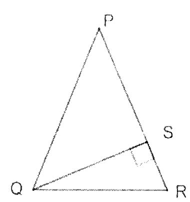
(Question 2)

7. For which of the following values of k the following equation is not satisfied?

(1) 3
(2) 4
(3) 5
(4) 7

(Question 4)

8. In Δ PQR, PQ = PR and S is a point on PR such that QS PR. Given that PS and SR are of integer length in cm and $QS^2 = 63$ cm², the least value of SR in cm is



(1) 7
(2) 1
(3) 3
(4) none of these

(Question 10)

9. Five friends Ahmed, Baskar, Charlie, Dravid, Easwar went to a lodge `Stay Safe' to seek room(s). The lodge manager Rahul said that there are only 3 rooms vacant of which two of them are double rooms - Room 143 and Room 145 and the other a single room - Room 131. The friends agreed to take up the three rooms and paid the advance amount for the same. [note: Exactly 2 persons can stay in a double room and Exactly 1 person can stay in a single room]. Here are few possibilities of 5 friends occupying the 3 rooms:

Serial number	Occupying *Room numbers*		
(Possibility)	143	145	131
1	A, B	C, D	E
2	A, C	B, D	E
3	C, D	A, B	E
4	E, B	D, A	C
⋮	⋮	⋮	⋮

Code: A – Ahmed; B – Baskar; C – Charlie; D – Dravid; E - Easwar

In how many ways can the friends occupy the 3 rooms (including the above cases)?

(1) 60
(2) 30
(3) 15
(4) 120

(Question 24)

10. A rectangle of length 27 cm and breadth 12 cm is dissected into two congruent hexagons.
These two hexagons can be reassembled to form a square. Then, the perimeter of each hexagon in cm is

(1) 60
(2) 56
(3) 48
(4) 45
(Question 26)

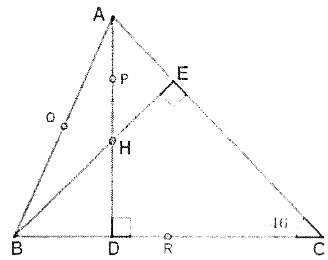
11. y is a 3-digit number such that it is equal to 16 times the sum of its digits. Total number of such 3-digit numbers y is

(1) 0
(2) 1
(3) 2
(4) 3
(Question 5)

12. Find the number of *integer* pairs (a, b) such that $ab - 3a + 4b = 0$.

(1) 4
(2) 6
(3) 8
(4) 12
(Question 6)

13. In the adjoining figure $\angle C = 46^o$, AD and BE are the altitudes cutting at H. P, Q, R, are respectively the mid points of HA, AB and BC. Then the measure of the angle PQR is



(1) 92°
(2) 98°
(3) 94°
(4) None of these

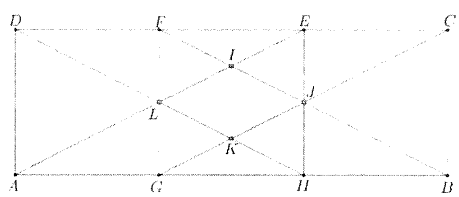
(Question 15)

14. In how many ways 60 can be divided into 10 different natural numbers?

(1) 8
(2) 12
(3) 10
(4) 7

(Question 21)

15. ABCD is a rectangle where AB = 3 cm; BC = 1 cm. G, H, are points on side AB such that AG = GH = HB, as shown. F, E are points on side DC such that CE = EF = FD, as shown. AE, BF intersect at point I and CG, BF intersect at point J, as shown. HD, CG intersect at point K and HD, AE intersect at point L, as shown. If the area of the rectangle ABCD is N times the area of the quadrilateral IJKL, then N is



(1) 10
(2) 18
(3) 12
(4) 16
(Question 28)

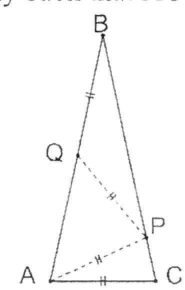
16. For how many values of K is 30^{15} the least common multiple of 20^5, 6^8, 15^5, 5^{15} and K?

(1) 10
(2) 15
(3) 20
(4) none of these
(Question 11)

17. y is a positive integer such that $11 + y$; $13 + y$; $17 + y$; $19 + y$ are all prime numbers.
The sum of the digits of the *smallest* such y is

(1) 10
(2) 9
(3) 8
(4) 7
(Question 27)

18. $AB = BC$ in an isosceles triangle ΔABC. Points P, Q are on sides BC,AB respectively such that $AC = AP = PQ = QB$, as shown. The value of is



(1) 4
(2) 6
(3) 8
(4) 10
(Question 25)

19. Let N be the number $y^2 - x^2$, with $x = 781,781,781...,7815$ having 2020 digits (781 repeating 673 times) and $y = 465346534653...46539$ having 2021 digits (4653 repeating 505 times). The remainder when N is divided by 22 is

(1) 12
(2) 14
(3) 16
(4) 10
(Question 39)

20. The sum to 50 terms of the following series where [x] is the greatest integer $\leq x$, is
(Note that the sum of the numerator and denominator of the fractions are increasing from 2 onwards. For sum two there is one term, for sum three two terms, for sum 4, three terms and so on.)

(1) 68
(2) 70
(3) 71
(4) 72
(Question 36)

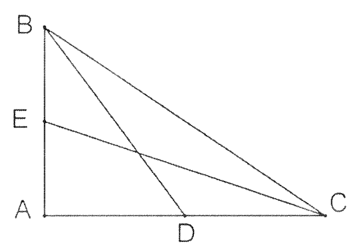
21. Let m and n be *positive integers* such that $2021^2 + m^2 = 2020^2 + n^2$. The sum of all possible values of $m + n$ where m, n satisfy the above equation is

(1) 4041
(2) 4490
(3) 5837
(4) 5850
(Question 13)

22. F is a simplest fraction whose numerator and denominator are both composite numbers that differ by 9. If < F < , then the largest possible sum of the numerator and denominator of F is

(1) 99
(2) 81
(3) 89
(4) 93
(Question 23)

23. Let ABC be a right triangle with $\angle A = 90^{\circ}$. The length of the medians through B and C are and . Then AB =



(1) 10
(2) 20
(3) 24
(4) 25

(Question 42)

24. The addition given below is incorrect. This can be made correct by changing one digit *d*, wherever it occurs into another digit e. Then is

$$\begin{array}{r} 6395427 \\ +\ 5547504 \\ \hline 13044031 \end{array}$$

(1) 30
(2) 12
(3) 20
(4) 42

(Question 45)

25. What are the last two digits of the *quotient* of 10^{50} when divided by $10^{25} + 7$?

(1) 07
(2) 49
(3) 93
(4) None of these
(Question 37)

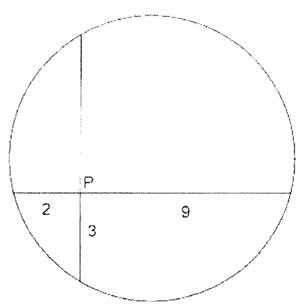
26 A, B, C, D are four numbers given. The sum of a set of three numbers from these are recorded as 1140, 1250, 1335, 955. Then the sum of the largest and smallest among the four numbers is

(1) 380
(2) 820
(3) 2290
(4) none of these
(Question 19)

27. Given that x, y, u, v are real numbers such that $x + y + u = 4$; $y + u + v = -5$; $u + v + x = 0$; $v + x + y = -8$. Then xyuv =

(1) 180
(2) 210
(3) 240
(4) 250
(Question 29)

28. Two perpendicular chords of a circle intersect at P. The lengths of the parts of the chords are given in the figure. Then double the square of the radius of the circle is ----



(1) 32.5
(2) 65
(3)
(4)
(Question 43)

29. ABC is a 3-digit number such that its last digit C is divisible by 2; the two digit part BC is divisible by 3; ABC itself is divisible by 4. The number of such 3 digit numbers is

(1) 81
(2) 72
(3) 63
(4) 54
(Question 46)

30. Let 2499, 3554 and 4398 leave the same remainder "r" when they are divided by $d > 1$. Then, the value of $(d - r)$ is

(1) 33
(2) 178
(3) 211
(4) None of these
(Question 38)